AUE-CSC 104: Computer Hardware (3 units)

Dr. A.R. Iyanda

Recommended Textbook and Materials

- Computer Organization And Architecture Designing For Performance Eighth Edition- William Stallings
- Computer Organization Design: THE HARDWARE/SOFTWARE INTERFACE- David A. Patterson and John L. Hennessy

Requirements

- Class participation and attendance
- Assessment (Assignment, Quiz, Test, Examination)
- Practical

Outline

- A survey of fundamentals with emphasis on hardware and systems concepts. Interfaces and configurations. Computer application modes: Batch processing: Process/operation control interactive, real Times. Timesharing. Multiplexing/ Demultiplexing: MODEMS, ADC, DAC, Systems Viability: Graceful degradation; MTTR, MTBF etc. Computer Selection for a given application, economic versus technical considerations.
- Computer circuits; diode arrays, PIAs etc, Integrated circuits fabrication process.Use of MSI, LSI and VLSI IC' hardware Design. Primary and Secondary memories; core memory, etc. Magnetic devices; disks, tapes, video disks etc. Peripheral devices; printers, CRT's, keyboards, character recognition. Operational amplifiers; Analog-to- digital and Digital-to-analog converter. Analog computers.

Definitions

- Hardware refers to the physical components of a computer system that are responsible for executing tasks and supporting the operations of software. These tangible devices work together to perform tasks such as processing data, storing information, providing input, and generating output.
- Hardware devices in a computer system include the Central Processing Unit (CPU), memory (RAM and storage devices), input/output devices (keyboard, monitor, mouse), and more.

Significance of Hardware in a Computer System

- Execution of Tasks: Hardware is the foundation upon which software runs. Without hardware, software cannot function, as it provides the necessary infrastructure to process and store data.
- Interaction with Software: Hardware and software work in tandem to deliver system functionality. Hardware executes the instructions provided by software (applications, programs) through processing units, memory, and I/O devices.
- Performance and Efficiency: The quality and capability of hardware impact system performance. For example, faster processors, more memory, and high-quality storage can significantly improve the overall speed and responsiveness of the system.
- Dependability: Properly configured and maintained hardware ensures system stability and longevity. It impacts a system's ability to handle workloads, prevent system failures, and provide consistent operation.

Distinction Between Hardware and Software

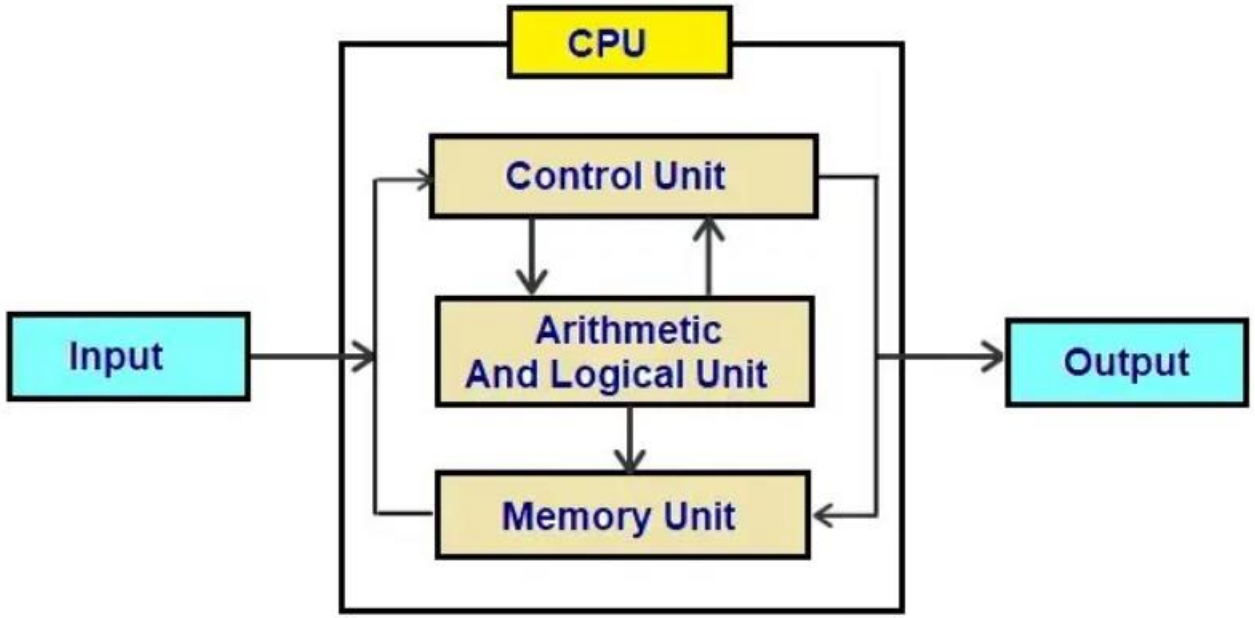
- Both hardware and software are essential components of a computer system, but they serve fundamentally different roles:
- 1.Hardware:
- Physical Components: Hardware refers to the actual physical part of the computer, directly involved in processing data, and can be touched and seen. Examples include the motherboard, CPU, RAM, hard drive, monitor, and keyboard.
- Direct Execution: Hardware executes the instructions provided by the software and performs tasks such as computation, data storage, and input/output operations.
- Characteristics:
  - Tangible, physical components
  - Requires physical space and energy to operate
  - Can be upgraded or replaced

Distinction Between Hardware and Software

- 2.Software:
- Virtual Components: Software refers to the programs, applications, and operating systems that run on the hardware. A collection of instructions and data. It provides instructions to the hardware on how to operate and interact with data.
- Indirect Interaction: Software tells the hardware what to do, and does not physically exist. Without software, the hardware remains idle. There are two types (System and Application software)
- Characteristics:
  - Intangible, non-physical code
  - Stored digitally on hardware (e.g., hard disk, cloud storage)
  - Can be updated or patched to improve functionality

Hardware Components

- A computer system consists of four primary categories of hardware components, each serving specific functions: input, output, processing, and storage.



Hardware Components

- Input Components: These are hardware devices used to enter data and commands into the computer. They provide a way for users to interact with the system.
- Examples:
  - Keyboard: Used for typing input commands and data.
  - Mouse: Allows user to point and select options on a screen.
  - Scanner: Digitizes physical documents and converts them into a digital format.
  - Microphone: Converts sound into data that can be processed by the computer.

Hardware Components

- Output Components: These hardware devices are used to present the results of computer processing to the user. The output can be visual, auditory, or hard copy.
- Examples:
  - Monitor: Displays images, text, and video output for the user.
  - Printer: Produces a physical copy of digital documents.
  - Speakers: Produce audio output, such as system alerts, music, or video sound.

Hardware Components

- Processing Components: The central components responsible for performing the majority of the computer's operations, including computation and control.
- Examples:
  - Central Processing Unit (CPU): Often referred to as the "brain" of the computer, the CPU interprets and executes instructions from programs via the control unit (CU) and performs calculations with the arithmetic and logic unit (ALU).
  - Graphics Processing Unit (GPU): A specialized processor designed to handle rendering images, videos, and visual data efficiently.

Hardware Components

- Storage Components: These devices store data and instructions that are needed during processing or for future use. Storage can be temporary (volatile/ primary) or permanent (non-volatile/ secondary).
- Examples:
  - Random Access Memory (RAM): A form of temporary memory used by the CPU to store and quickly access data and instructions during active operations.
  - Hard Disk Drives (HDDs): Permanent storage devices that store data even when the computer is turned off.
  - Solid-State Drives (SSDs): Similar to HDDs but with faster data access speeds as they use flash memory to store data.
  - Optical Discs (CD/DVD/Blu-ray): Storage devices that use lasers to read and write data onto a disk.
  - USB Drives and SD Cards: Portable storage devices commonly used for transferring and saving data.

Hardware Components

- There is also a Bus System which comprises of:
  - Data Bus: Transports data between components of the computer.
  - Address Bus: Carries addresses to and from the memory.
  - Control Bus: Carries control signals to coordinate the activities of the system.

Basic Working Principle of a Computer

- A computer follows a systematic process to perform tasks. The working principle can be described as the Fetch-Decode-Execute Cycle, also known as the Machine Cycle.
- Fetch: The control unit fetches (retrieves) an instruction from memory (RAM) based on the Program Counter (PC), which points to the address of the next instruction.
- Decode: The instruction fetched is decoded by the control unit. The decoding process translates the instruction into signals that the CPU understands, preparing it for execution.
- Execute: The ALU executes the operation defined by the instruction (e.g., addition, subtraction, logical operations).
- Store (if needed): After execution, the results may be stored back into memory (RAM) or into a register for further processing.

Types of Computers

- Computers come in different types based on their use, size, capabilities, and processing power.
- Desktop Computers: a general-purpose computer that is typically designed to be stationary, as it requires external peripherals like a monitor, keyboard, and mouse.
- Components:
  - Tower or all-in-one setup with CPU, memory, storage, and expansion slots.
  - Large displays and multiple peripheral connections.
- Typically used for office tasks, gaming, multimedia applications, software development, etc.

- Laptop Computers: A portable version of a computer that integrates a monitor, keyboard, storage, and battery into a compact form, allowing for mobility.
- Components: Similar to desktops but in a smaller, integrated, and lightweight design, with a built-in battery.
- Designed for portability and often used by individuals, professionals, and students for a wide range of tasks such as browsing, programming, multimedia, and gaming.

Types of Computers

- Servers: powerful computers that manage, store, and process data for other computers (clients) over a network. Servers are typically designed for specific tasks such as hosting websites, databases, and applications.
- Components: High-capacity CPUs, RAM, storage, and sometimes, specialized components like GPUs for handling large-scale workloads.
- Use for Web hosting, data storage, cloud computing, enterprise applications, and databases.

- Embedded Systems: Embedded systems are specialized computers designed to perform specific tasks within larger systems. They are usually small and operate on a limited set of functions, often without direct user interaction.
- Components: Microcontrollers or microprocessors, limited memory, and sometimes, real-time operating systems (RTOS).
- Found in everyday devices like smartphones, cars, home appliances (e.g., washing machines, microwave ovens), medical devices (e.g., heart monitors), and industrial machinery.

Interfaces and Configurations

- An interface in computing is a point of interaction between two systems, devices, or software components, enabling communication and data exchange. It allows one system to communicate with another, ensuring that inputs and outputs are appropriately processed.
- Types of Interfaces:
  - Hardware Interface: The physical and logical connections between devices (e.g., USB ports, network adapters).
  - Software Interface: Communication between different software programs (e.g., Application Programming Interfaces - APIs).
  - Human-Machine Interface (HMI): The system through which users interact with the computer (e.g., graphical user interfaces - GUI, command-line interfaces - CLI).

Hardware Interfaces

- Peripheral Interfaces
- Universal Serial Bus (USB): A widely used interface for connecting peripherals such as keyboards, mice, printers, external storage devices, etc.
- USB Versions:
  - USB 2.0: Up to 480 Mbps.
  - USB 3.0: Up to 5 Gbps.
  - USB 3.1/3.2/USB-C: Higher data transfer speeds and reversible connections.
- Function: Both for data transfer and providing power to connected devices.

Hardware Interfaces

- Network Interfaces
- Ethernet: A standard networking interface for wired connections between computers and devices, typically using twisted pair cables and TCP/IP protocol. Commonly used in local area networks (LANs) for connecting devices such as computers, printers, and routers.
- Wi-Fi: A wireless networking interface allowing devices to connect to local area networks using radio waves.
- Bluetooth: A short-range wireless interface for connecting devices like keyboards, headsets, and smartphones to computers.

Hardware Interfaces

- Display Interfaces
- HDMI (High-Definition Multimedia Interface): Transmits high-definition video and audio between devices like computers, monitors, TVs, and projectors. HDMI 1.4 (supports 1080p resolution), HDMI 2.0 (supports 4K resolution).
- VGA (Video Graphics Array): An older interface for video output, commonly found on projectors and older monitors.

Software Interfaces

- Application Programming Interface (API)
- An API provides an interface for software applications to communicate with each other. It defines functions, commands, data structures, and objects that developers can use to interact with the software without needing to understand the implementation details.
- Example: Google Maps API that allows third-party developers to integrate map functionality into their own applications.

Software Interfaces

- Graphical User Interface (GUI)
- A GUI enables users to interact with computers through graphical icons, buttons, and menus rather than text commands.
- Examples: Windows OS, macOS, Android.

- Command-Line Interface (CLI)
- A CLI allows users to input text commands for program execution. It is faster and more powerful but less user-friendly compared to GUIs.
- Examples: Bash (Linux), Command Prompt (Windows).

Configurations

- Configuration refers to the arrangement and settings of hardware and software within a system, defining its behavior and functionality. Proper configuration can ensure that hardware components and software run optimally. Type of configuration are:
  - Hardware Configuration
  - Network Configuration
  - Software Configuration
  - Security Configuration

Configurations

- Hardware Configuration
- CPU Configuration: Settings affecting CPU performance (e.g., overclocking settings, cores enabled).
- Memory Configuration: Choices between different types of RAM, speed settings (e.g., DDR4 or DDR5, frequency settings).
- Disk Configuration: How disks are arranged for performance, such as in RAID (Redundant Array of Independent/Inexpensive Disk) setups
- BIOS/UEFI Configuration: Basic settings that control how the computer boots and interacts with peripherals.

Configurations

- Network Configuration
- IP Addressing: Assigning static or dynamic IP addresses to network devices.
  - Static IP: Manually assigned for fixed devices such as servers.
  - Dynamic IP (DHCP): Automatically assigned by a DHCP (Dynamic Host Configuration Protocol) server (for devices like smartphones and laptops).
- DNS Settings: Configuring the Domain Name System to resolve domain names to IP addresses for accessing websites.
- Subnet Mask: Divides an IP address into network and host parts, crucial for network segregation

Configurations

- Software Configuration
- Operating System Settings: Various configuration options such as user management, access control, and system performance optimizations.
  - For example, adjusting power settings on Windows to high performance mode or battery saver mode.
- Application Settings: Customizing the behavior and features of applications.
  - Example: Adjusting security settings in an antivirus program or defining storage paths in cloud applications.

Configurations

- Security Configuration
- Firewall Settings: Configuring firewalls to allow or block specific network traffic (e.g., blocking port 80 for HTTP access).
- Encryption: Configuring encryption algorithms and keys for securing data, whether for email, hard drives, or communication.

Assignment

- Define and explain the significance of hardware in a computer system.
- Describe the four basic hardware components of a computer system and their roles.
- Discuss the differences between (i) SSD and HDD as storage devices (ii) RAM and ROM?
- What is the function of a CPU, and how does it contribute to the processing of data in a computer?
- Discuss various Interfaces and Configurations that exist